1 *Monitor CPU Utilization and Memory Utilization for detecting unauthorized process activations. (Hint: More CPU utilization as compared to Memory is an indicator of anomaly)

b. Create complete memory dump using windows

c. Read Memory Dump Using Windows Driver toolkit

Answer -

Aim

To monitor CPU and memory utilization for detecting unauthorized process activations, to create a complete memory dump using Windows, and to analyze the memory dump using Windows Driver Toolkit.

(a) Monitoring CPU Utilization and Memory Utilization

Monitoring CPU and memory utilization is an important technique in Digital Forensics to detect unauthorized or suspicious processes running in a system.

CPU Utilization

- CPU utilization refers to the percentage of processing power used by active processes.
- It indicates how busy the CPU is.
- It is measured in percentage (%).
- It can be monitored using Task Manager.

Abnormal conditions:

- Sudden high CPU usage
- Unknown processes consuming CPU
- Continuous high usage without user activity

Causes:

- Malware
- Cryptojacking
- Unauthorized background processes

Memory Utilization

- Memory utilization refers to the amount of RAM used by applications and processes.
- It is measured in MB or GB.
- It shows memory consumption.
- It can be monitored using Task Manager.

Abnormal conditions:

- Very high memory usage
- Memory leaks
- Suspicious background programs

Detection of Unauthorized Processes

- High CPU usage with low memory usage indicates an anomaly.
- Malware often uses high CPU but less memory.
- Normal programs use both CPU and memory proportionally.
- Sudden spikes in CPU indicate suspicious activity.

(b) Creating Complete Memory Dump Using Windows

A memory dump is a file that stores the contents of RAM at a specific time and is used in digital forensics for analysis.

Types of Memory Dump

- Complete Memory Dump
- Kernel Memory Dump
- Small Memory Dump
- Complete memory dump stores entire RAM and is best for forensic analysis.

Steps to Create Complete Memory Dump

1. Open Control Panel
2. Go to System and Security → System
3. Click Advanced System Settings
4. Under Startup and Recovery, click Settings
5. In “Write debugging information”, select Complete Memory Dump
6. Set location (default: C:\Windows\MEMORY.DMP)
7. Click OK and restart system

Manual Method

- Press Ctrl + Scroll Lock twice
- System crashes and creates dump file

Uses of Memory Dump

- Captures running processes
- Detects hidden malware
- Helps in debugging
- Provides forensic evidence

(c) Reading Memory Dump Using Windows Driver Toolkit

Windows Driver Toolkit is used to analyze memory dump files. The main tool used is WinDbg.

Steps to Read Memory Dump

1. Install Windows Driver Toolkit (WDK)
2. Open WinDbg
3. Click File → Open Crash Dump
4. Select file: C:\Windows\MEMORY.DMP

5. Load symbols using:
    - .symfix
    - .reload
6. Run command:
    - !analyze -v

Information Obtained

- Running processes
- System errors
- Loaded drivers
- Crash details
- Suspicious activity

Forensic Importance

- Helps detect malware behavior
- Identifies unauthorized processes
- Useful in cyber investigation
- Provides detailed system analysis

Conclusion

Monitoring CPU and memory utilization is an effective method for detecting unauthorized processes. High CPU usage with low memory usage indicates suspicious activity. Creating and analyzing memory dumps using tools like WinDbg plays a crucial role in digital forensics for identifying system threats, analyzing system behavior, and ensuring overall system security.

2 *Study the DFRWS Investigative Model and apply it in a simulated digital forensic investigation (Consider digital forensic scenario like a case involving a potential data breach or unauthorized access to a computer system).

a. Investigate according to phases of model.

b. Prepare report detailing the steps taken during the investigation.

Answer -

Aim

To study the DFRWS investigative model and apply it in a simulated digital forensic investigation involving unauthorized access or data breach, and to prepare a detailed investigation report.

(a) Investigation According to Phases of DFRWS Model

The **DFRWS Investigative Model** is used to conduct systematic digital forensic investigations. It consists of the following phases:

Scenario

An organization suspects that an employee has gained unauthorized access to a system and leaked confidential data.

---

1. Identification Phase

- Identify the incident and confirm whether unauthorized access has occurred.
- Detect suspicious activities such as unusual login attempts.
- Identify affected systems, users, and data.
- Example: Multiple failed login attempts followed by successful login at unusual time.

---

2. Preservation Phase

- Preserve digital evidence to avoid tampering or loss.
- Disconnect affected system from network.
- Create forensic image of storage devices.
- Maintain integrity of evidence and chain of custody.

---

3. Analysis Phase

- Analyze collected data to understand the incident.
- Examine logs, files, and user activities.
- Identify how the unauthorized access occurred.
- Create timeline of events.
- Example: User accessed confidential files and transferred them externally.

---

4. Documentation Phase

- Document all steps taken during investigation.
- Record tools used, methods applied, and evidence collected.
- Maintain proper records for legal purposes.
- Ensure documentation is clear and detailed.

---

5. Presentation Phase

- Present findings in a structured and understandable format.
- Summarize the investigation results.
- Provide evidence and conclusions.
- Prepare report suitable for legal or organizational use.

(b) Report of the Investigation

Title: Digital Forensic Investigation Report

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1. Introduction

This report describes the investigation of a suspected unauthorized access and data breach using the DFRWS model.

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2. Objective

- To identify unauthorized access
- To collect and analyze digital evidence
- To determine the cause and impact of the incident

---

3. Methodology

The investigation was conducted using the phases of the DFRWS model:

- Identification
- Preservation
- Analysis
- Documentation
- Presentation

---

4. Steps Taken During Investigation

Identification:

- Suspicious login activity detected
- Unauthorized access identified

Preservation:

- System isolated from network
- Data preserved to prevent changes

Analysis:

- Logs and files analyzed

- Unauthorized data access confirmed
- Timeline of events created

Documentation:

- All steps recorded
- Evidence and tools documented

Presentation:

- Findings summarized
- Report prepared

---

5. Findings

- Unauthorized user accessed system
- Confidential data was accessed
- Evidence found in system logs

---

6. Conclusion

The investigation confirms that unauthorized access occurred and sensitive data was compromised. The DFRWS model provided a structured approach for conducting the investigation and analyzing the evidence effectively.

---

7. Recommendations

- Use strong authentication methods
- Regularly monitor system logs
- Implement access control mechanisms
- Use security tools like intrusion detection systems

3. Analyse a real-world or hypothetical case where ethical issues arose in a digital forensics investigation Task to be performed by students:

a. Select a real-world case of a digital forensics investigation where ethical issues played a significant role (e.g., the case of the FBI's investigation of the San Bernardino iPhone, The Ashley Madison Hack (2015))

b. Analyse the case based on following points: Ethical issues involved in the investigation Situation handling procedure followed by Investigator Does the investigation base on professional ethical norms Or what Ethical guidelines should be followed

c. Prepare Report on ethical issues, their impact on the investigation and a conclusion on how the situation could have been managed ethically

Answer -

Aim

To analyze a real-world case where ethical issues arose in a digital forensic investigation and evaluate their impact along with proper ethical practices.

(a) Selection of Case

The selected case is the **Sony Pictures Hack**.

In 2014, **Sony Pictures Entertainment** was targeted by hackers who gained unauthorized access to its systems and leaked confidential data such as employee information, emails, and unreleased movies.

(b) Analysis of the Case

1. Ethical Issues Involved

- **Privacy Violation:**
  Personal data of employees was exposed publicly.
- **Confidential Information Leak:**
  Sensitive business data and unreleased content were leaked.

- **Misuse of Data:**
  Leaked emails were used to harm individuals and company reputation.
- **Ethical Responsibility of Investigators:**
  Investigators had to handle sensitive data carefully.

---

2. Situation Handling Procedure Followed by Investigators

- Cybersecurity teams started investigation after the breach.
- Systems were secured to prevent further damage.
- Digital forensic experts analyzed logs and hacked data.
- Efforts were made to identify attackers.
- Government agencies were involved in investigation.

---

3. Whether Investigation Followed Professional Ethical Norms

- Investigation followed proper legal procedures.
- Evidence was collected and analyzed systematically.
- However, leaked data caused ethical concerns regarding privacy.
- Investigators needed to ensure sensitive data was not misused.

---

4. Ethical Guidelines That Should Be Followed

- Maintain confidentiality of sensitive data
- Protect privacy of employees
- Use data only for investigation purposes
- Avoid unnecessary data exposure
- Follow professional ethical standards

(c) Report on Ethical Issues and Their Impact

Title: Ethical Analysis Report of Digital Forensic Investigation

---

1. Introduction

This report analyzes the ethical issues involved in the Sony Pictures Hack and its impact on digital forensic investigation.

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2. Description of Case

In 2014, hackers attacked Sony Pictures and gained unauthorized access to confidential data. Large amounts of sensitive information were leaked, causing major damage to the organization.

---

3. Ethical Issues Identified

- Exposure of personal employee data
- Leakage of confidential business information
- Damage to individual privacy
- Ethical challenges in handling leaked data

---

4. Impact on Investigation

- Increased difficulty in protecting sensitive information
- Reputational damage to the company
- Ethical pressure on investigators
- Need for careful handling of digital evidence

---

5. Ethical Evaluation

- The attack violated privacy and confidentiality principles
- Investigators had to balance evidence collection with data protection
- Ethical handling of sensitive data was necessary

---

6. Proper Ethical Approach

- Ensure strict data confidentiality
- Limit access to sensitive information
- Avoid public exposure of private data

- Follow ethical and legal standards
- Protect affected individuals

Conclusion

The Sony Pictures Hack highlights the importance of ethics in digital forensic investigations. The exposure of sensitive data caused serious harm and raised ethical concerns regarding privacy and confidentiality. Investigators must follow strict ethical guidelines, protect sensitive information, and ensure responsible handling of digital evidence to maintain trust and ensure justice.

4 *Investigate data in a cloud environment, focusing on issues like data privacy and security breaches

a. Conduct a forensic analysis of cloud storage (e.g. Dropbox, Google Drive) for potential data breaches or misuse

b. Retrieve access logs and analyse activities that suggest unauthorized access or tampering (Hint: Use Cloud storage APIs, AWS Cloud Trail, Google Cloud Platform logs.)

Answer -

Aim

To investigate data in a cloud environment and analyze potential data breaches and unauthorized access using forensic techniques.

Objective

- To understand cloud forensic investigation
- To identify data privacy and security issues
- To analyze access logs for suspicious activities
- To detect unauthorized access or data tampering

(a) Forensic Analysis of Cloud Storage

Cloud storage platforms such as **Google Drive** and **Dropbox** are widely used to store data, making them important in digital forensic investigations.

Steps

1. Select a cloud platform (Google Drive / Dropbox)
2. Log in using an authorized or test account
3. Examine stored files and folders carefully
4. Collect file metadata such as:
    - File name

    - Date created and modified
    - File size
5. Check for suspicious activities:
    - Unknown or unfamiliar files
    - Unexpected file modifications
    - Unauthorized sharing links
    - Missing or deleted files

Data Privacy and Security Issues

- Sensitive data may be exposed to unauthorized users
- Improper sharing settings can lead to data leakage
- Weak authentication increases risk of unauthorized access
- Lack of monitoring may allow unnoticed data breaches

(b) Retrieval and Analysis of Access Logs

Access logs provide detailed information about user activities and help detect unauthorized access.

Steps

1. Open activity or history section of the cloud platform
2. Retrieve logs including:
    - Login timestamps
    - Device information
    - IP address or location
    - File access and modification details
3. Analyze logs to identify:
    - Unknown login locations
    - Unusual login times
    - Multiple failed login attempts

    - Unauthorized file access or deletion
    - Repeated suspicious activities

Tools Used

- **AWS CloudTrail** for tracking user activity
- **Google Cloud Platform** logs for monitoring access
- Cloud APIs for retrieving and analyzing log data

Output

- Cloud platform analyzed: Google Drive / Dropbox
- Access logs retrieved successfully

Observations:

- Login detected from unknown location
- File modified without user knowledge

Identified Issues:

- Unauthorized access
- Suspicious file activity
- Possible data breach

Conclusion

Cloud forensic investigation is useful for detecting unauthorized access and data breaches. By analyzing stored data and access logs, suspicious activities can be identified and prevented. Proper monitoring, strong authentication, and secure access control are important to protect cloud data.

5 Collect live data on Windows\Linux:

a. Create a response toolkit on windows having utilitycmd.exe, PsLogged On, netstat

b. Establish TCP connection between forensic workstation and the target system using netcat

c. Run trusted cmd.exe, identify logged users and remote access users, Record creation, access times and all the modifications made to the files

Answer -

Aim

To collect and analyze live system data from a Windows/Linux system using forensic tools and identify active users, network connections, and file activities.

Objective

- To understand live data acquisition in digital forensics
- To create a response toolkit
- To establish TCP connection using Netcat
- To identify logged-in and remote users
- To analyze file creation, access, and modification

Procedure

Part A: Create Response Toolkit (Windows)

1. Create a folder named **Response Toolkit** on USB or external storage.
2. Add trusted forensic tools:
    - cmd.exe (trusted command prompt)

    - PsLoggedOn (to identify logged-in users)
    - netstat (to view network connections)
3. Ensure tools are verified and not tampered.
4. Use external storage to avoid modifying the target system.

Part B: Establish TCP Connection Using Netcat

Netcat is used to establish communication between systems.

1. On forensic workstation, open command prompt.
2. Start listening on a port:

nc -lvp 4444

3. On target system, open command prompt.
4. Connect to forensic workstation:

nc <forensic_IP> 4444

5. Verify that TCP connection is successfully established.

Part C: Live Data Collection and Analysis

1. Run trusted cmd.exe from response toolkit.
2. Identify logged-in users:

PsLoggedOn.exe

    - Displays local and remote users
3. Check active network connections:

netstat -an

    - Shows active and listening connections
4. Identify remote access sessions:
    - Look for established external connections
    - Detect suspicious IP addresses
5. Analyze file activity:
    - Check file properties such as:

        - Created time
        - Modified time
        - Accessed time
    - Alternatively, use command:

      dir /T:C /T:A /T:W

6. Record all observations carefully for forensic documentation.

Output

- List of logged-in users obtained
- Active network connections identified
- Remote access sessions detected

Observations:

- Suspicious remote connection found
- Unknown user session detected
- File modification activity observed

Conclusion

Live data collection is an important step in digital forensics as it allows investigators to capture real-time system information without shutting down the system. Using trusted tools like Netcat, PsLoggedOn, and netstat, investigators can identify users, network activity, and file changes effectively, helping in detecting unauthorized access and preserving crucial evidence.

6 Create Forensic Images with any Imager Tool like Exterro FTK Imager

Answer -

Aim:

To create a forensic image of a storage device using FTK Imager for digital forensic investigation while preserving data integrity.

Objective:

- To understand the process of forensic imaging
- To create an exact bit-by-bit copy of digital evidence
- To use FTK Imager tool effectively
- To verify integrity using hash values

Procedure:

Step 1: Install and Launch Tool

1. Download and install FTK Imager from the official website
2. Open the FTK Imager application

Step 2: Create Disk Image

1. Click on **File → Create Disk Image**
2. Select source type:
   o Physical Drive
   o Logical Drive
   o Image File
3. Choose the required drive and click **Next**

Step 3: Select Image Format

1. Choose image type:
   o E01 (recommended and compressed)
   o RAW (dd format)
2. Click **Next**

Step 4: Enter Evidence Details

1. Add information such as:
   o Case Number
   o Evidence Number
   o Examiner Name
   o Description
2. Click **Next**

Step 5: Set Destination

1. Select destination folder
2. Enter image file name
3. Set fragment size if required
4. Enable **Verify images after acquisition**
5. Click **Finish**

Step 6: Acquire and Verify Image

1. Click **Start** to begin imaging
2. Wait for the process to complete
3. Verify generated hash values (MD5/SHA1)

Output:

• Forensic image created successfully (E01/RAW format)
• Hash values generated and verified
• Evidence details recorded

Conclusion:

Forensic imaging allows investigators to create an exact copy of data without altering the original evidence. Using FTK Imager ensures accurate acquisition and maintains data integrity through hash verification, making it reliable for forensic investigations.

7 *Perform Hashing to verify the authenticity of digital evidence

a. Create a file and generate a hash (MD5, SHA-256) using hashing tools

b. Alter the file slightly and generate the hash again to observe how the hash changes (Use HashCalc, MD5 & SHA Checksum Utility, Pythons hashlib or any such tool)

Answer -

Aim:

To verify the integrity and authenticity of digital evidence by generating and comparing hash values.

Objective:

- To generate hash values using MD5 and SHA-256
- To observe changes in hash values after modifying a file
- To compare original and modified hash values
- To detect data tampering

Introduction:

Hashing is a technique used to convert data into a fixed-length value called a hash. It helps in verifying whether a file has been changed. Even a small change in the file results in a completely different hash value, making it useful for maintaining data integrity.

Procedure:

Part A: Create File and Generate Hash

1. Create a text file (e.g., sample.txt).
2. Add some content and save the file.
3. Open Command Prompt (Windows) or Terminal (Linux).
4. Generate MD5 hash:

Windows:

certutil -hashfile sample.txt MD5

Linux:

md5sum sample.txt

5. Generate SHA-256 hash:

Windows:

certutil -hashfile sample.txt SHA256

Linux:

sha256sum sample.txt

6. Note down both hash values.

Part B: Modify File and Recheck Hash

1. Open the same file again.
2. Make a small change (add a word or space).
3. Save the file.
4. Generate MD5 and SHA-256 hash again using the same commands.
5. Record the new hash values.
6. Compare them with the previous values.

Observation:

• Original file produced specific hash values
• Modified file produced completely different hash values
• Even a small change resulted in a major difference

Output:

• MD5 and SHA-256 hash values generated successfully
• Hash values changed after modification
• File integrity verified

Conclusion: Hashing is used to verify the integrity of digital evidence. Any small change in a file results in a different hash value, helping to detect tampering.

8 Recover deleted or corrupted files from a storage device and perform file carving (e.g., photos, documents) using any data recovery tool• Multilevel inheritance.

Answer -

Aim:

To recover deleted or corrupted files from a storage device and perform file carving using a data recovery tool.

Objective:

• To understand data recovery in digital forensics
• To recover deleted or corrupted files
• To perform file carving for extracting data
• To analyze recovered files such as photos and documents

Introduction:

Data recovery is an important process in digital forensics used to retrieve deleted or corrupted files from storage devices. Even after deletion, data may remain on the disk until it is overwritten.

File carving is a technique used to recover files based on file signatures without relying on file system structure. It helps in recovering files such as images, documents, and videos.

Procedure:

Part A: Recover Deleted Files

1. Connect the storage device (USB/Hard Disk) to the system.
2. Open a data recovery tool such as:
    - Recuva
    - PhotoRec

3. Select the target drive for scanning.
4. Choose file types to recover (images, documents, etc.).
5. Start the scanning process.
6. Wait for the scan to complete.
7. Preview the recoverable files.
8. Select required files and click on Recover.
9. Save recovered files to a different location.

Part B: Perform File Carving

1. Open file carving tool (e.g., PhotoRec).
2. Select the storage device or disk image.
3. Choose file formats to recover:
    - JPG, PNG (images)
    - DOC, PDF (documents)
4. Start file carving process.
5. Tool scans raw data and extracts files based on signatures.
6. Recovered files are stored in output folder.

Multilevel Inheritance (Concept)

- Data recovery follows multiple levels:
  • Physical level (raw disk data)
  • Logical level (file system)
  • Application level (file formats)
- File carving works at physical level when file system is damaged.

Observation:

- Deleted files were successfully recovered
- Some corrupted files were partially recovered
- File carving extracted files without file names
- Recovered files included images and documents

Output:

- Deleted files recovered successfully
- Corrupted files partially restored
- File carving completed
- Images and documents retrieved

Conclusion:

Data recovery and file carving are essential techniques in digital forensics for retrieving deleted or damaged files. These methods help investigators recover important evidence even when the file system is corrupted or data is deleted.

9 *Read and Interpret Operating Systems logs on Windows file system Hint: Check whether the log gives information about file systems. Any such entry indicates some malicious activity

Answer -

Aim:

To read and analyze Windows operating system logs and identify suspicious or malicious activities related to the file system.

Objective:

- To understand Windows logging mechanisms
- To analyze system and security logs
- To identify file system-related events
- To detect suspicious or unauthorized activities

Procedure:

Step 1: Open Event Viewer

1. Press **Windows + R**
2. Type eventvwr.msc and press Enter
3. Event Viewer window will open

Step 2: Navigate Logs

1. Go to **Windows Logs**, which contains:
   o Application
   o Security
   o System
2. Focus on:
   o **Security Log** (login, access, user activity)
   o **System Log** (file system, disk, hardware events)

Step 3: Analyze File System Logs

1. Look for events related to:
   o File creation, deletion, and modification
   o Disk errors or file system corruption
2. Check important details:
   o Event ID
   o Date and Time
   o Source
   o Description

Step 4: Identify Suspicious Activity

1. Look for:
   o Unauthorized file access
   o Multiple failed login attempts
   o Sudden file deletions or modifications
   o Disk or NTFS-related errors
2. Use **Filter Current Log** option to find specific events easily

---

Step 5: Record Observations

1. Note any unusual or suspicious entries
2. Record Event IDs and details
3. Export logs or take screenshots if required

Output:

• System and security logs accessed successfully
• File system-related events identified
• Suspicious or abnormal activities recorded with Event IDs

Conclusion:

Windows logs provide important information about system and file activities. By analyzing logs using Event Viewer, investigators can detect unauthorized access, file system changes, and potential malicious activities, making it an essential tool in digital forensics.

10 Install Kali Linux Hint: Students can install Kali Linux on VMware Workstation/Virtual Box

Answer -

Aim:

To install Kali Linux on a virtual machine using VMware Workstation or VirtualBox.

Objective:

• To understand installation of Kali Linux
• To learn virtualization concepts
• To install Kali Linux using a virtual machine
• To configure system settings for proper working

Procedure:

Step 1: Download Required Software

1. Download **Kali Linux** ISO file from official website
2. Download and install virtualization software:
    - **VMware Workstation**
    - or **Oracle VM VirtualBox**

Step 2: Create Virtual Machine

1. Open VMware Workstation / VirtualBox
2. Click on **Create New Virtual Machine**
3. Select installation type (Typical/Custom)
4. Choose **ISO image file** of Kali Linux
5. Select operating system type as **Linux → Debian (64-bit)**

Step 3: Configure Virtual Machine

1. Allocate system resources:
    - RAM: Minimum 2 GB (recommended 4 GB)
    - Storage: Minimum 20 GB
2. Set processor cores as required
3. Configure network settings (NAT or Bridged)

Step 4: Install Kali Linux

1. Start the virtual machine
2. Select **Graphical Install** option
3. Choose language, region, and keyboard layout
4. Configure username and password
5. Partition the disk (Guided – use entire disk)
6. Wait for installation to complete

Step 5: Complete Installation

1. Install GRUB boot loader when prompted
2. Restart the virtual machine
3. Remove ISO file if required
4. Log in using created credentials

Step 6: Verify Installation

1. Check system is running properly
2. Open terminal and run basic commands
3. Verify internet connectivity

Output:

- Kali Linux installed successfully on virtual machine
- System configured and running properly
- User able to access terminal and tools

Conclusion:

Kali Linux can be successfully installed using virtualization software like VMware or VirtualBox. It provides a secure environment for learning cybersecurity and penetration testing tools without affecting the host system.

11 *Use Nmap utility to perform following tasks:

a. Install Nmap on Linux or Windows OS

b. Detect which devices are live on your local network. Identify the services and their versions running on a particular host

c. Detect the operating system of a target host

d. Perform a port scan on a specific set of ports

e. Perform an aggressive scan to gather as much information as possible about a target host

f. Use Nmap's scripting engine to search for vulnerabilities in a target system.

Answer-

Aim:

To use Nmap utility to perform network scanning tasks such as host discovery, service detection, OS detection, port scanning, aggressive scanning, and vulnerability scanning.

Objective:

- To install and use Nmap tool
- To detect live hosts in a network
- To identify services and their versions
- To detect operating system of a host
- To perform port scanning and aggressive scanning
- To identify vulnerabilities using Nmap scripts

Procedure:

(a) Install Nmap on Linux or Windows

1. Install **Nmap** on the system.
    - Windows: Download installer and follow setup
    - Linux:

sudo apt install nmap

2. Verify installation:

nmap --version

This confirms that Nmap is installed successfully.

(b) Detect Live Devices and Services

1. To identify active hosts in the network (host discovery), use:

nmap -sn 192.168.1.0/24

This scan shows which devices are live on the network.

2. To detect services and their versions (service enumeration), use:

nmap -sV 192.168.1.1

This identifies running services and their versions on the target host.

(c) Detect Operating System

1. To identify the operating system of a target host, use:

nmap -O 192.168.1.1

This scan attempts OS detection based on network behavior.

(d) Perform Port Scan

1. To scan specific ports on a target system, use:

nmap -p 21,22,80,443 192.168.1.1

This command checks whether selected ports are open, closed, or filtered.

(e) Perform Aggressive Scan

1. To gather detailed information about a target host, use:

nmap -A 192.168.1.1

This performs aggressive scanning including OS detection, version detection, and script scanning.

(f) Use Nmap Scripting Engine (NSE)

1. To scan for vulnerabilities using Nmap scripts, use:

nmap --script vuln 192.168.1.1

This detects known vulnerabilities in the target system.

Observation:

- Live hosts detected in network
- Services and versions identified
- OS details obtained
- Open ports detected
- Vulnerabilities identified

Output:

- Active devices listed successfully
- Services and versions displayed
- Operating system detected
- Port scan results obtained
- Vulnerability scan completed

Conclusion:

Nmap is a powerful network scanning tool used for host discovery, service detection, OS identification, and vulnerability scanning. It helps in analyzing network security and identifying potential threats effectively.